Secure Grade: Top-secret( ) Secret( ) Internal( ) Public(√)

RV1126B EVB1 EVB User Guide

（福州硬件开发中心）

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Preface

Overview

This document mainly introduces the basic functions and hardware features of RV1126B EVB1, the multi-functional hardware configuration, and the operation and usage methods of software debugging, aiming to help debuggers use RV1126B EVB1 faster and more accurately, and become familiar with the Pre-recording application scheme developed by the RV1126B chip.

Product version

The product versions corresponding to this document are as follows:

Product name	Product version
RV1126B EVB1	RK_EVB1_RV1126B_DDR4P216SD6R_V11_20250430
SC850SL camera modules	RK_EVB1_EXT_RV1126B_Sensor_SC850SL_V10_20250214GXL
Camera modules conversion card	RK_EVB1_EXT_RV1103B_SENSOR_CONVERT_40P_TO_24P_V10_20240806

Intended Audience

This guide is mainly intended for:

- Hardware development engineers
- Layout engineers
- Technical support engineers
- Test engineers

Revision History

This revision history recorded description of each version, and any updates of previous versions are included in the latest one.

Version No.	Author	Revision Date	Revision Description	Remark
V1.0	LZX	2025.6.17	First release	

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1. System Overview

1.1. RV1126B Introduction

RV1126B is a high-performance vision processor SoC for machine vision application, especially for AI related application.

It is based on quad-core ARM Cortex-A53 64-bit core which integrates NEON and FPU. There is a 32KB I-cache, and 32KB D-cache for each core and 512KB unified L2 cache. The build-in NPU supports INT8/INT6 hybrid operation and computing power is up to 3TOPS. In addition, with its strong compatibility, network models based on a series of frameworks such as TensorFlow/MXNet/PyTorch/Caffe can be easily converted.

RV1126B introduces a new generation totally hardware-based maximum 12-Megapixel ISP and post processor. It implements a lot of algorithm accelerators, such as HDR, 3A, LSC, 3DNR, 2DNR, sharpening, dehaze, fisheye correction, gamma correction, feature points detection and so on. A maximum 8-Megapixel AI-ISP is also introduced as a complement to traditional ISP, which offers superior spatial denoising performance and enhanced image enhancement effects.

1.2. RV1126B Diagram

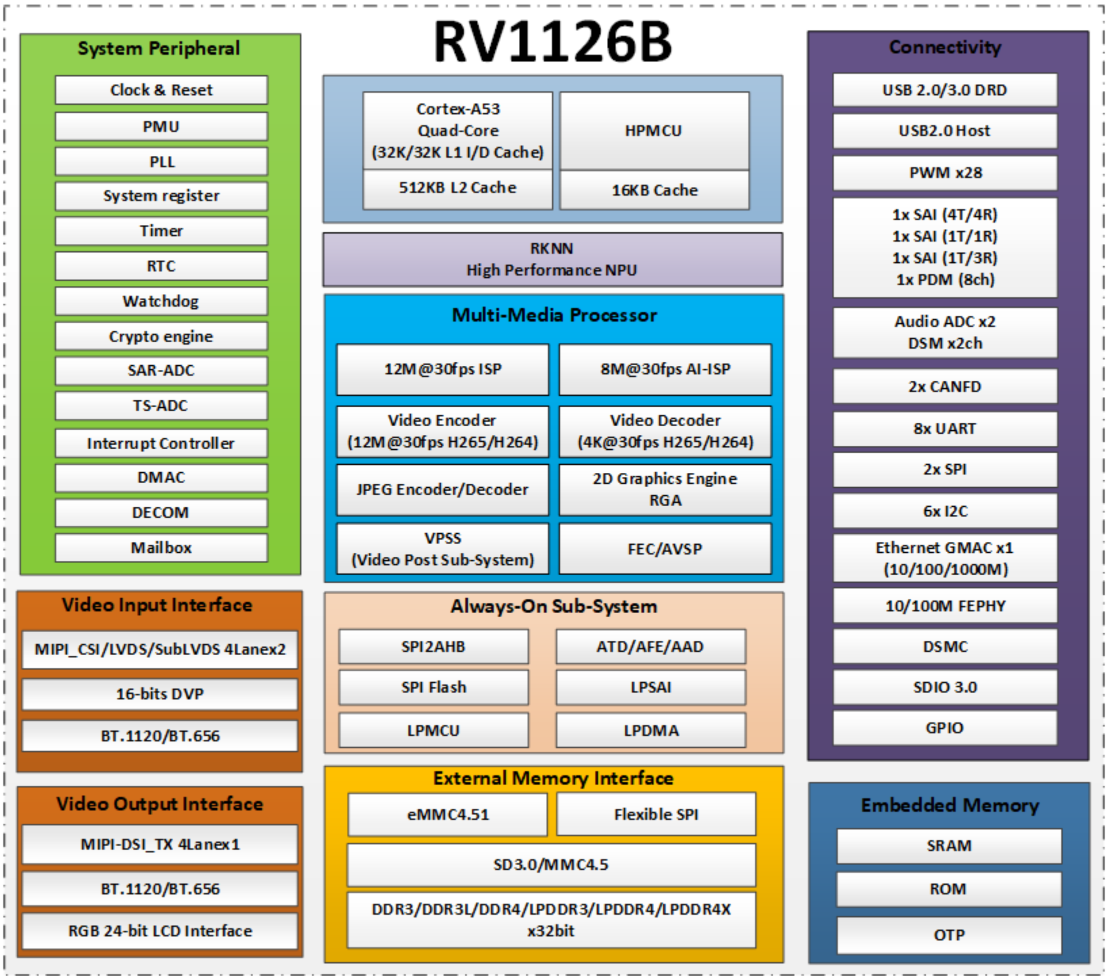


Figure 1-1 RV1126B chip diagram

1.3. System Introduction

1.3.1. System Diagram

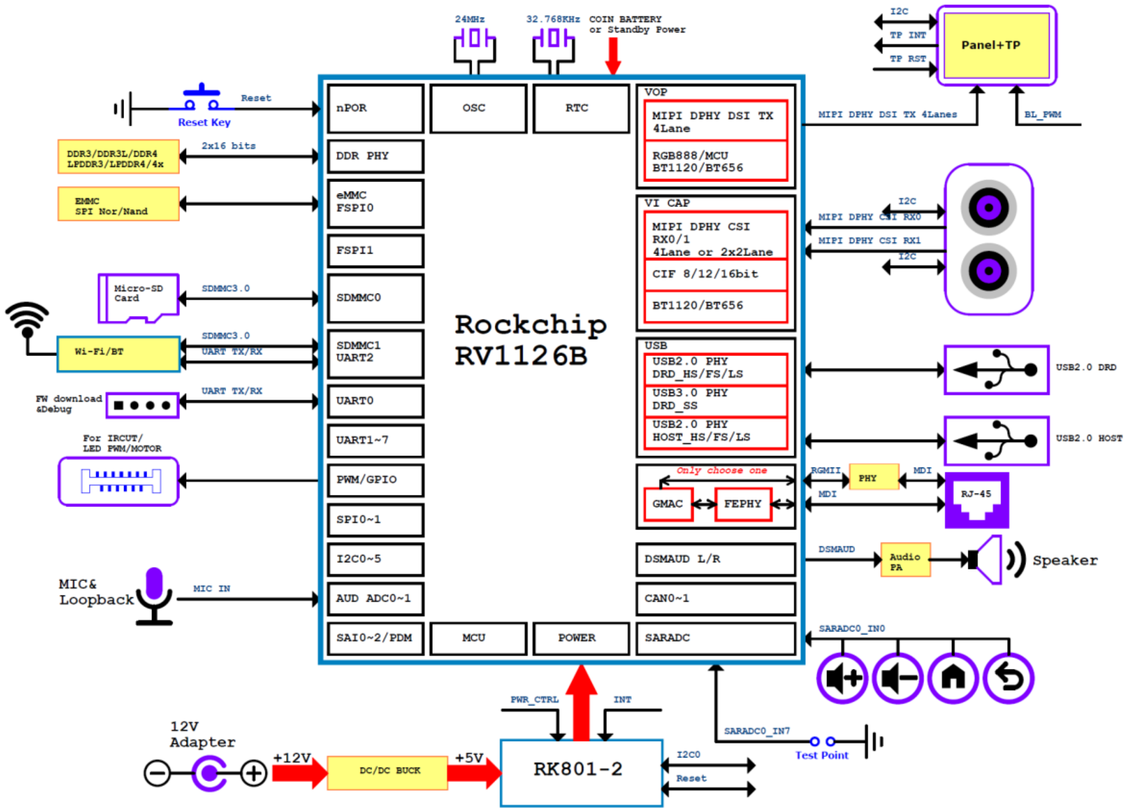


Figure 1-2 RV1126B EVB1 application sample

1.3.2. Function Overview

The functions included in the RV1126B EVB1 are as follows:

- Power supply: Interface for DC 12V adapter power supply.
- Video input interface: Supports 2x4Lane/4x2Lane MIPI input methods. Users can design the adapter board to achieve multi-camera input. EVB1 natively supports 4Lane monocular input.
- Video output interface: supports MIPI screen
- Network interface: Supports 1xRJ45 interface for 10/100Mbps Ethernet. Supports external expansion of a Gigabit Ethernet port via RGMII, which is led out through a 61082-60pin socket.
- Audio interface: Supports speaker output and dual MIC recording;
- Wi-Fi: Supports SDIO WIFI;
- UART Debug: Users debug and view LOG information for use; Support Type-C interfaces;
- JTAG: System JTAG debugging interface;
- System Key: Includes UPDATE, PWRON, RIGHT, LIFT, MENU, ESC/RECOVERY key
- RTC: Supports real-time clock and can be powered by a development board or a button battery (CR1220-3V);

1.3.3. Function Modules Distribution

RV1126B EVB1 Function interface distribution diagram:

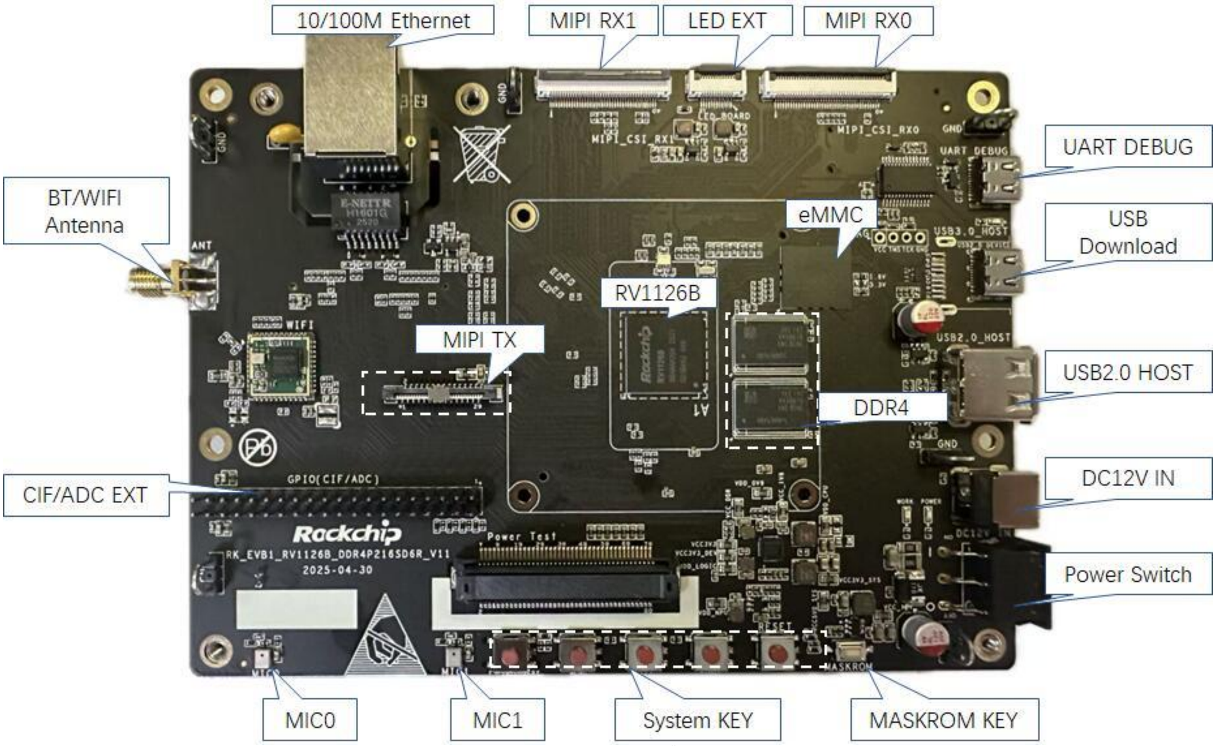


Figure 1-3 RV1126B EVB1 function interface distribution (front)

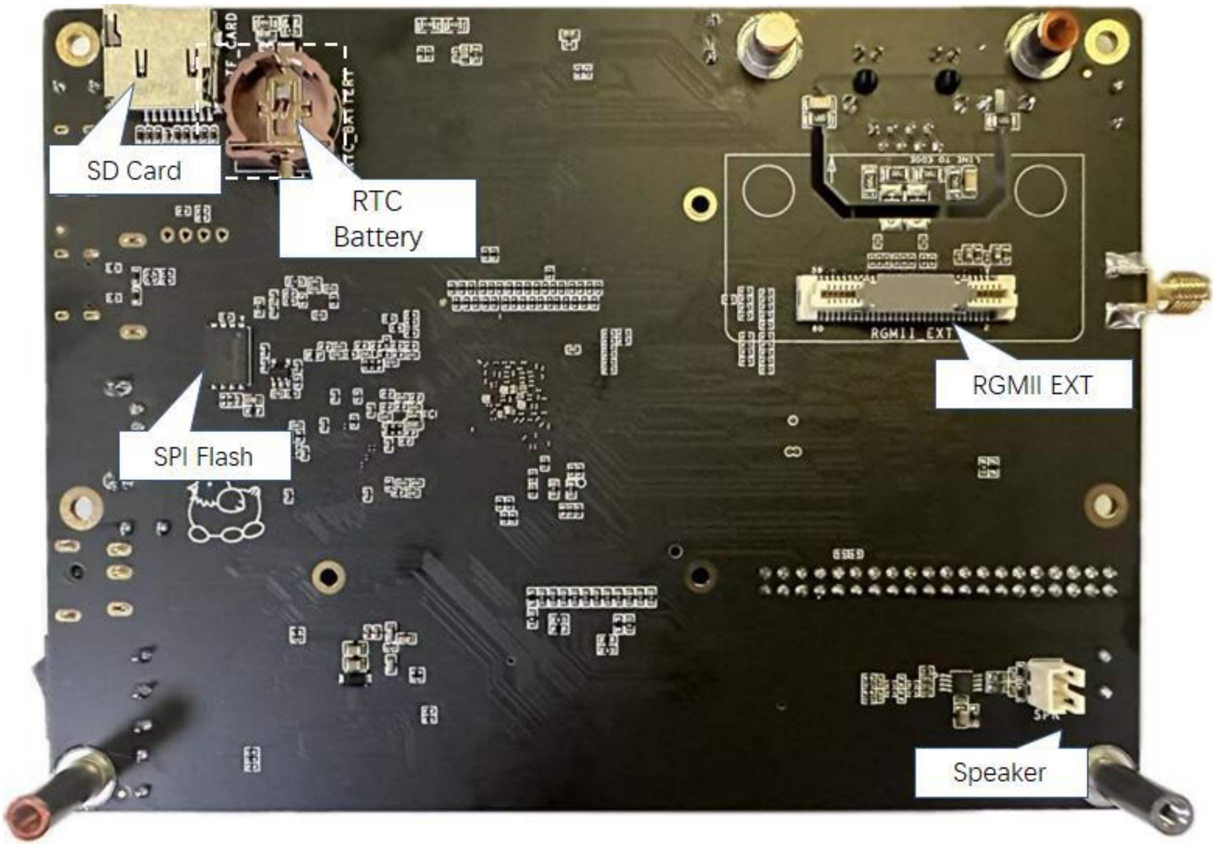


Figure 1-4 RV1126B EVB1 function interface distribution (back)

1.4. Components

The RV1126B EVB1 kit includes the following items:

- RV1126B EVB1
- Power adapter, default specifications: Input 100V AC~240V AC, 50Hz; Output 12V DC, 3A
- SC850SL monocular lens module
- 1080P MIPI display screen
- One 2.4G single-band SMA male connector antenna

2. Hardware Introduction

2.1. The Pictures

Figure 2-1 RV1126B EVB1

2.2. Power Block Diagram

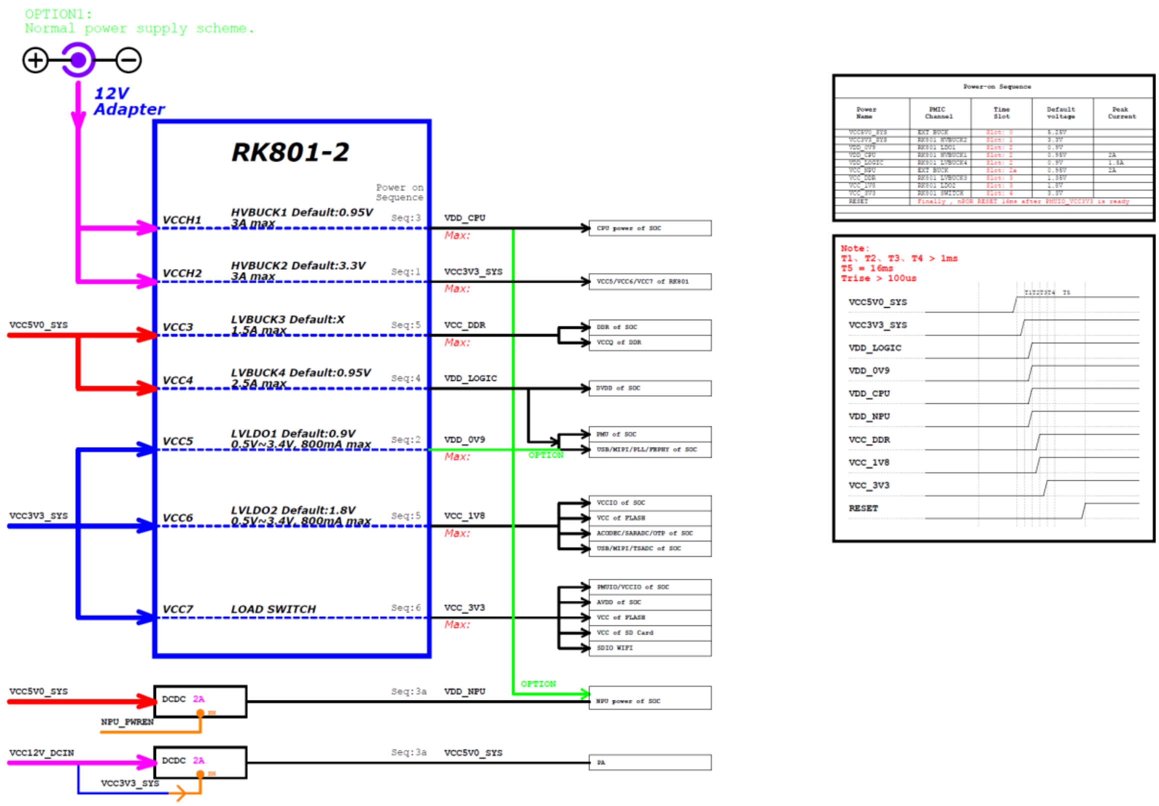


Figure 2-2 RV1126B EVB1 Block diagram

2.3. Extension Connector Information

During actual use, users may create extension boards. The connector model for the development board is as follows:

The U4700, U4800 are horizontal single-row 40-pin connection socket with 0.15mm pins and 0.5mm spacing. The dimensions are as follows:

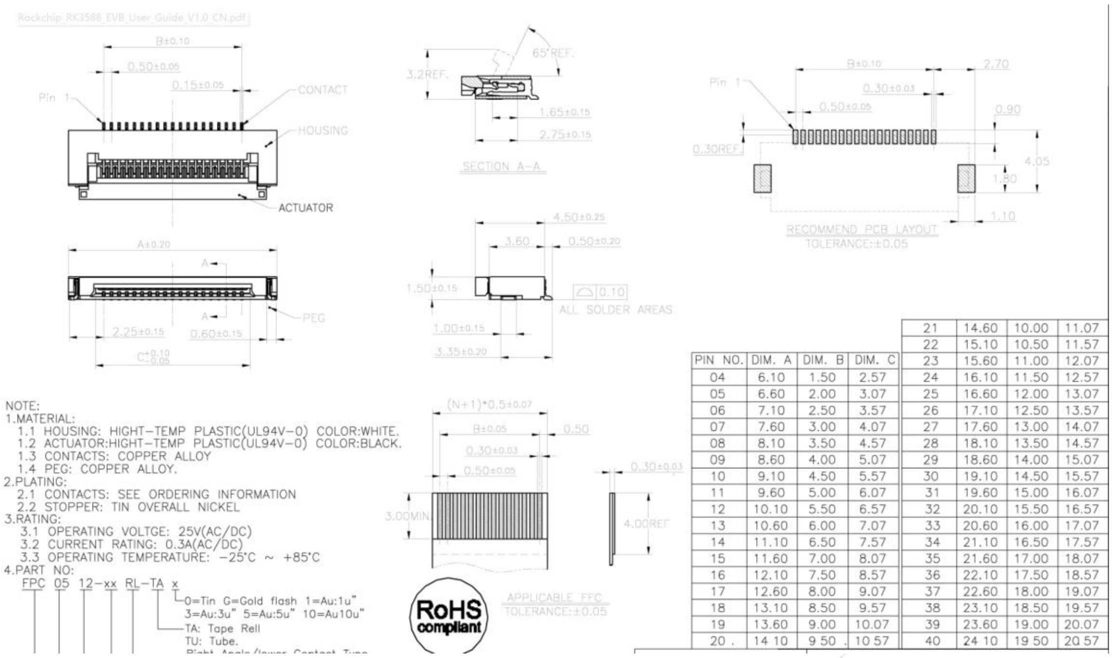
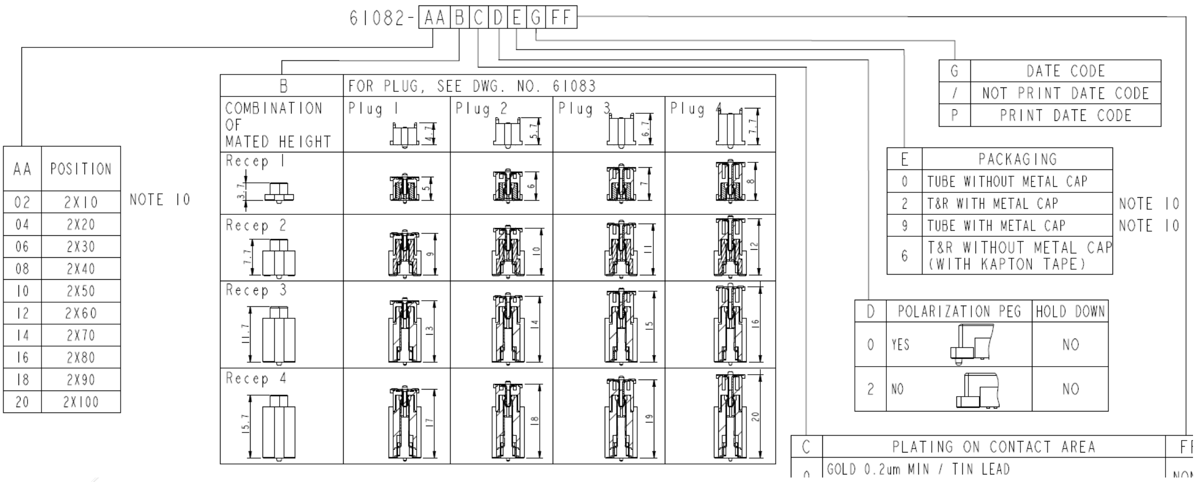
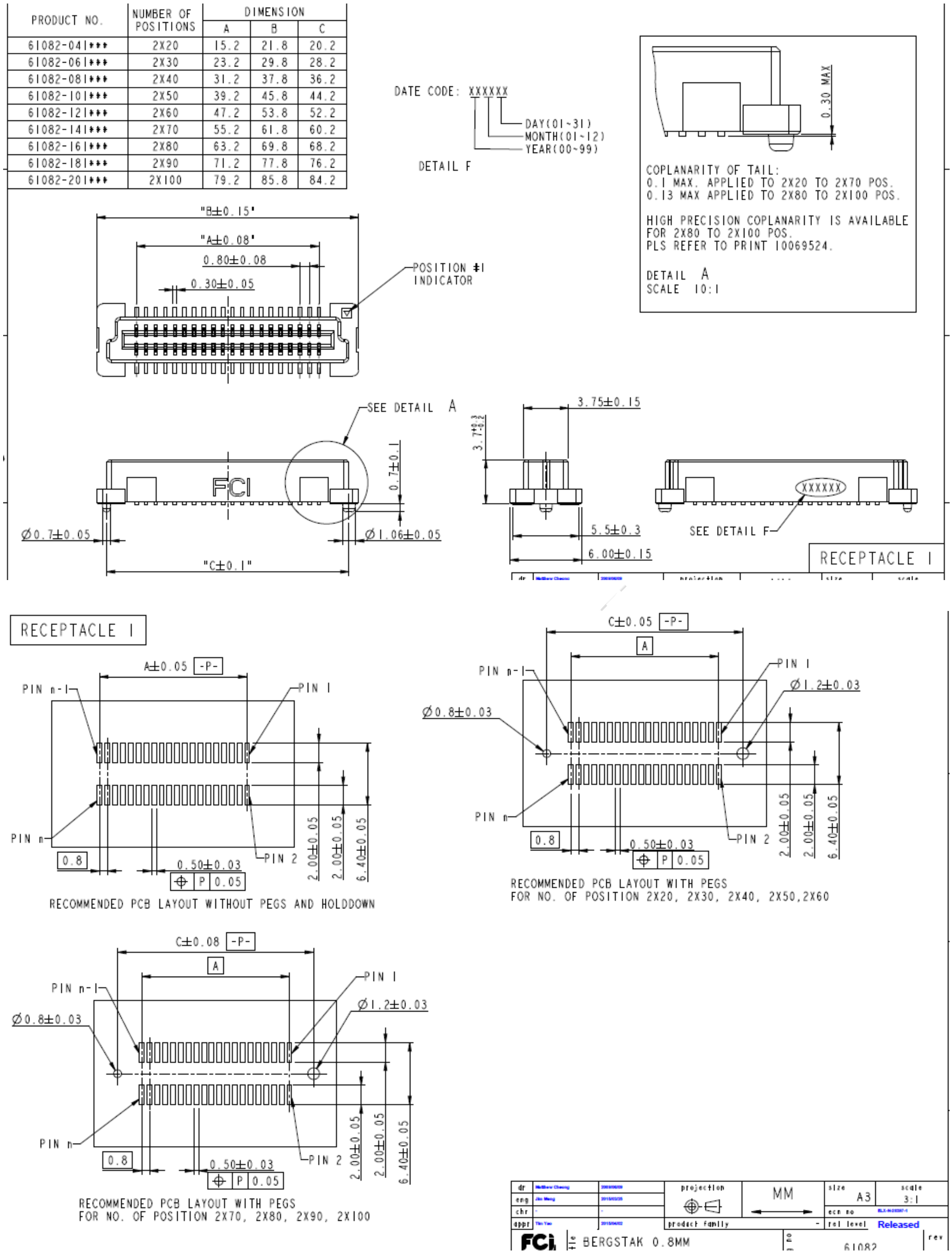


Figure 2-3 RV1126B EVB1 extension board connector

If need RGMII extension board, the connector model of the EVB1 is 61082-061402LF and the corresponding ext-board connector is 61083-061402LF. The size information is as follows.

The J6600 is a vertical double-row 60PIN connector with a pin width of 0.5mm and a pitch of 0.8mm. The specific dimensions are as follows:





PRODUCT NO.	NUMBER OF POSITIONS	DIMENSION A	B	C
61082-041***	2X20	15.2	21.8	20.2
61082-061***	2X30	23.2	29.8	28.2
61082-081***	2X40	31.2	37.8	36.2
61082-101***	2X50	39.2	45.8	44.2
61082-121***	2X60	47.2	53.8	52.2
61082-141***	2X70	55.2	61.8	60.2
61082-161***	2X80	63.2	69.8	68.2
61082-181***	2X90	71.2	77.8	76.2
61082-201***	2X100	79.2	85.8	84.2

Figure 2-4 J6600 Connector

2.4. Reference Diagram

The reference diagram and PCB design information corresponding to the EVB1 are as follows:

- Schematic: RK_EVB1_RV1126B_DDR4P216SD6R_V11_20250430.DSN
- PCB: RK_EVB1_RV1126B_DDR4P216SD6R_V11_20250430_final.brd

The SC850SL Camera module is used to debug 8M cameras. The reference diagram and PCB design information are as follows:

- Schematic: RK_EVB1_EXT_RV1126B_Sensor_SC850SL_V10_20250212.DSN
- PCB：RK_EVB1_EXT_RV1126B_Sensor_SC850SL_V10_20250214GXL.brd

The Camera Adapter Board corresponding to the EVB1 is used to connect the EVB and the module board. The corresponding reference diagram and PCB version information are as follows:

- Schematic: RK_EVB1_EXT_RV1103B_Sensor_Convert_40p_to_24p_V10_20240806.DSN
- PCB: RK_EVB1_EXT_RV1103B_SENSOR_CONVERT_40P_TO_24P_V10_20240806.brd

3. Module Overview

3.1. Power Supply

The power adapter inputs a 12V/3A power supply. After being stepped down by the high - voltage PMIC, it outputs different power supplies for the development board to use.

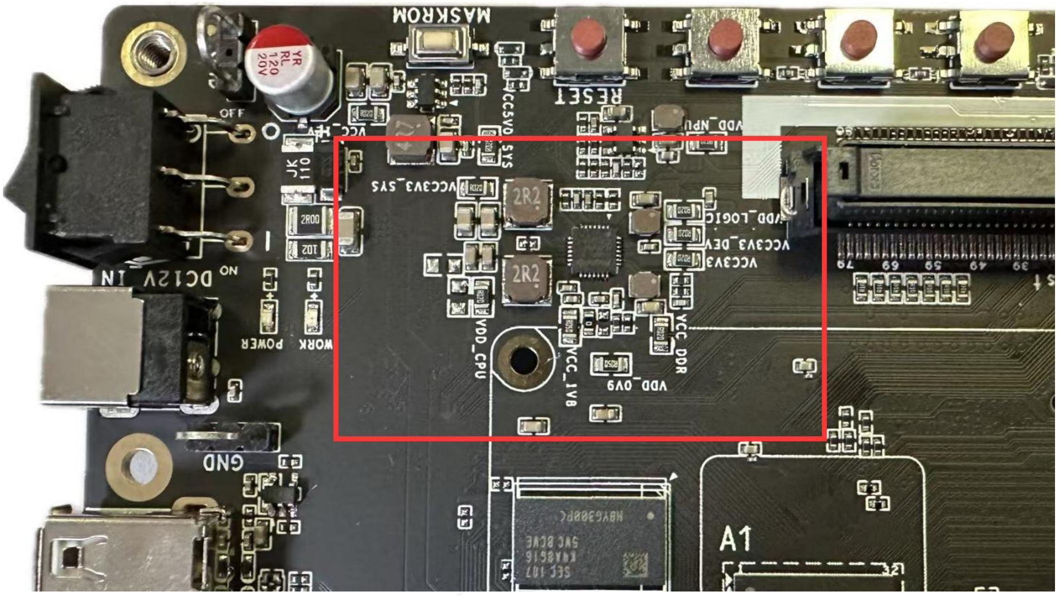


Figure 3-1 RV1126B EVB1 Power Input

3.2. Memory

There are two types of memory on the RV1126B EVB1: SPI Flash and eMMC Flash, with eMMC being the default.

If SPI Flash is to be used, please note that the FLASH power supply configuration needs to be modified to 1.8V/3.3V(EVB1 default by 3.3V), and the SPI FLASH signal selection resistor show in Figure 3-3 needs to be mounted and disconnecting the signal selection resistor for eMMC FLASH.

- eMMC: The memory type on the development board is eMMC Flash, with a default capacity of 32GB.
- SPI Flash: The development board is reserved for SPI Flash device.

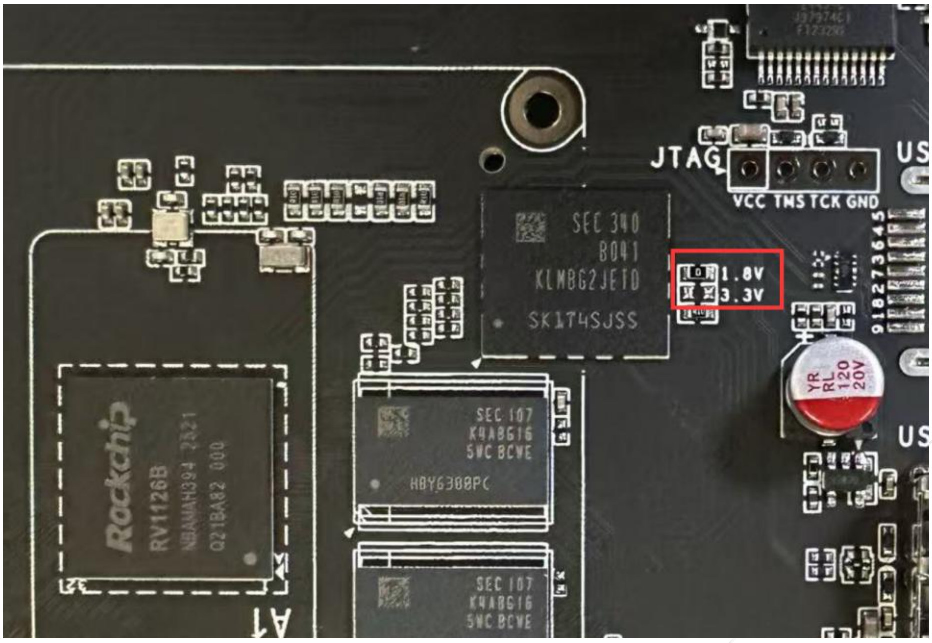


Figure 3-2 RV1126B EVB1 Flash IO level modify

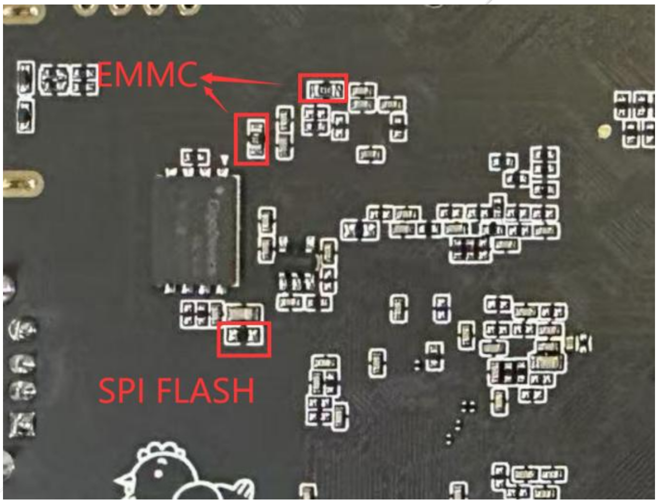


Figure 3-3 RV1126B EVB1 EMMC/SPI FLASH power selection

3.3. RTC

The RV1126B EVB uses the built-in RTC of the chip, which be powered by a on-board CR1220-3V button battery (when power is off), ensuring the continuous provision of an accurate real-time clock.

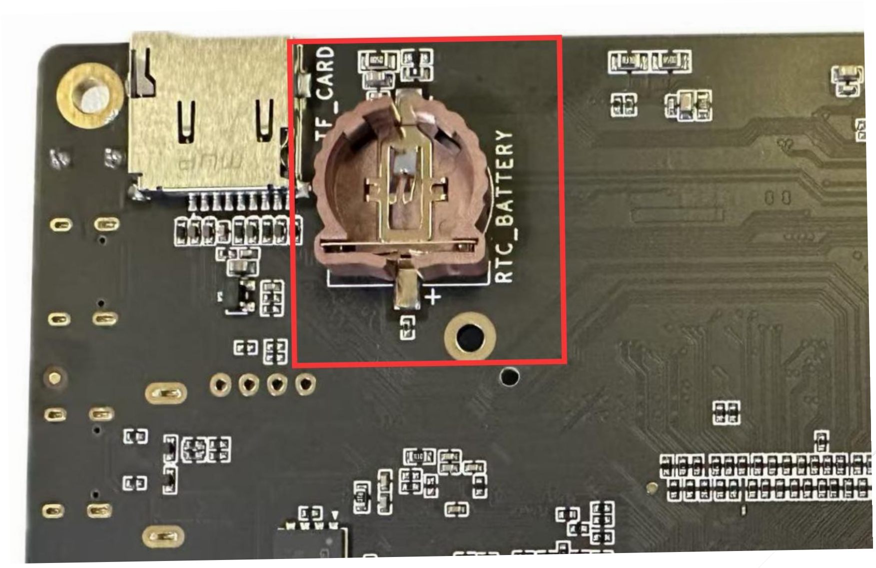


Figure 3-4 RV1126B EVB1 RTC Battery Holder

3.4. Keys

The development board uses SARADC_IN0 (ESC/RECOVERY, MENU, LEFT, RIGHT) as the key detection port, supporting a 13-bit resolution. The 4 ADC KEY keys can be configured for various purposes.

Press the MASKROM key while powering on to enter the programming mode.

The key locations are as follows:

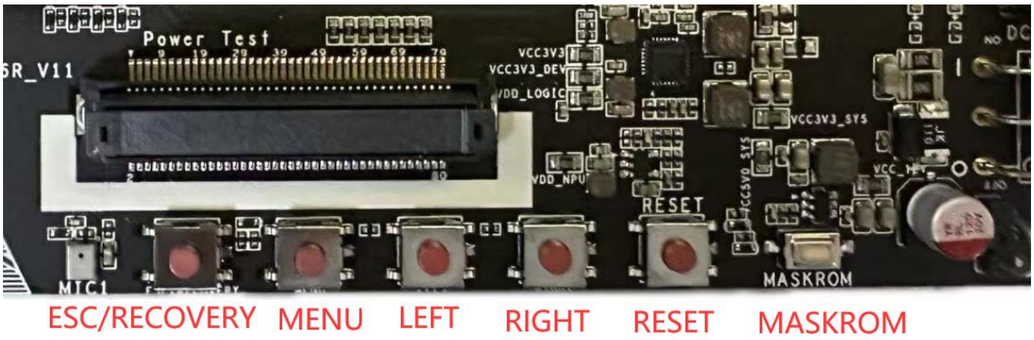


Figure 3-5 RV1126B EVB1 Keys

3.5. Ethernet Port

The development board supports the RJ45 interface and can provide 10/100Mbps Ethernet connection functions. The features are as follows:

- Compatible with the IEEE802.3 standard, supporting full-duplex and half-duplex operations, as well as cross-detection and auto - negotiation.
- Supports data rates of 10/100Mbps.
- The interface uses an RJ45 interface with indicators.

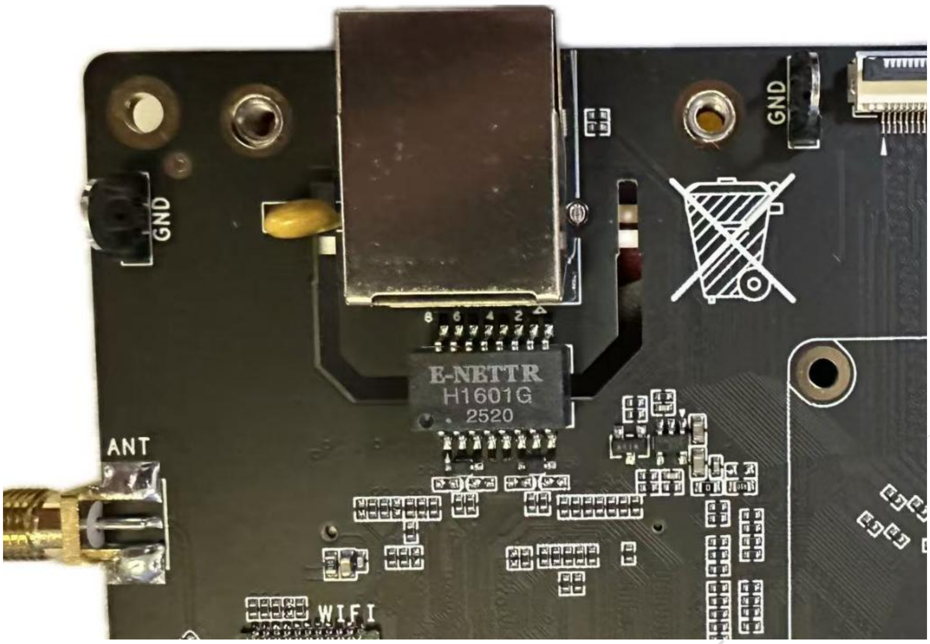


Figure 3-6 RV1126B EVB1 3.5. Ethernet Port

3.6. WiFi Interface

The development board uses RK'S WiFi module, with the following characteristics:

- Based on the RK960 chip.
- Supports BT5.3.
- Supports 1x1 WIFI (2.4G, 802.11 b/g/n) with an external SMA antenna interface.
- Powered by 1.8/3.3V(3.3V by default).
- Supports SDIO2.0/SDIO3.0.

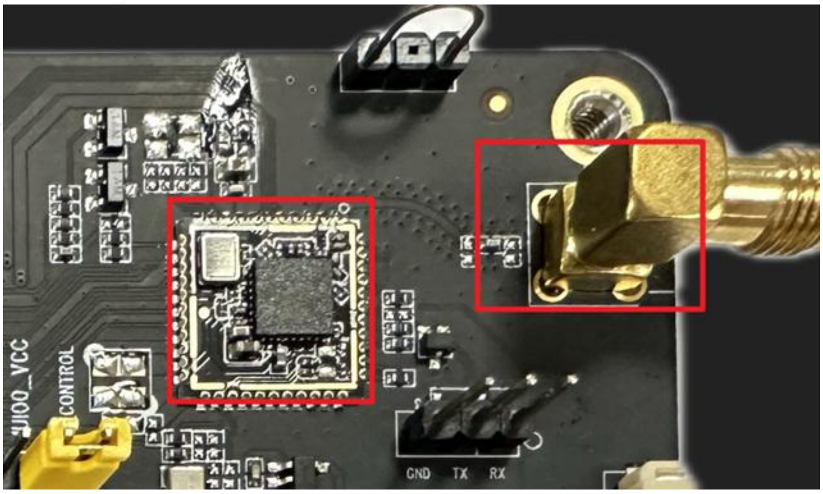


Figure 3-7 RV1126B EVB1 WiFi interface

3.7. UART Debugging Interface

The development board supports a USB TYPEC serial debugging interface.

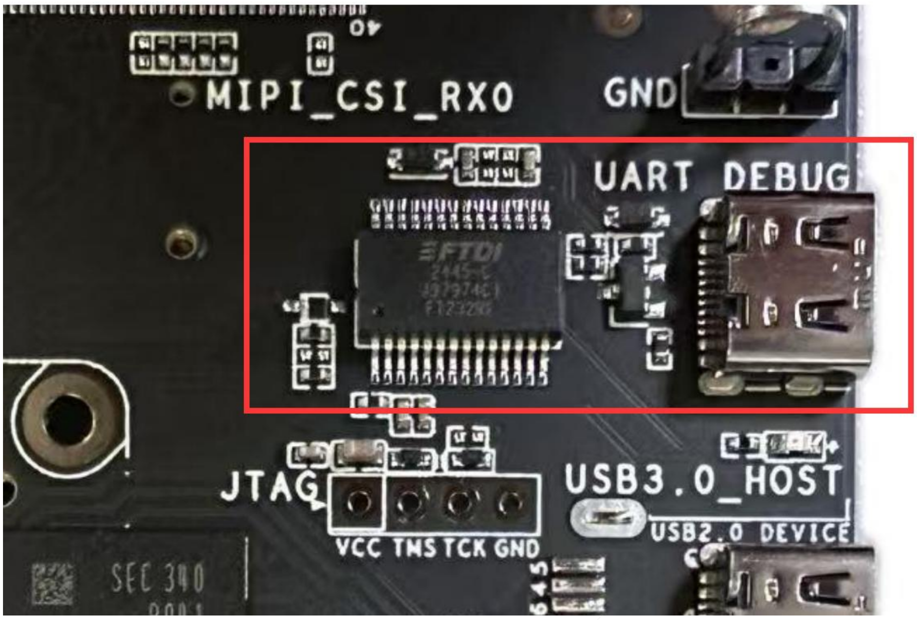


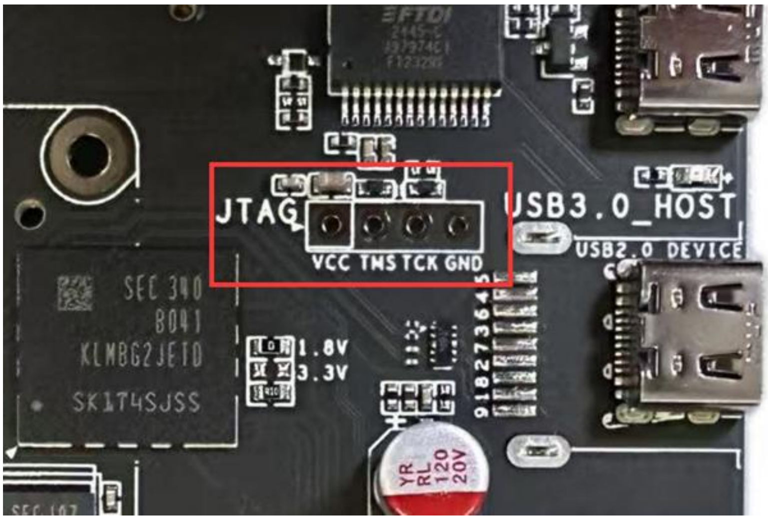
Figure 3-8 RV1126B EVB1 UART Debugging Interface

3.8. JTAG Debugging Interface

The development board reserves one JTAG interface. The JTAG interface of RV1126B is multiplexed with the SDMMC interface. Therefore, this interface is used as the JTAG debugging port by default when no TF card is inserted. When a TF card is inserted, it will be forced to switch to TF card mode, and the JTAG mode will be unavailable.

The JTAG socket supports ARM/HPMCU/LPMCU JTAG.

If need to use the JTAG interface, you need to solder the 2.54mm pin header and the resistors R9815 and R9816.



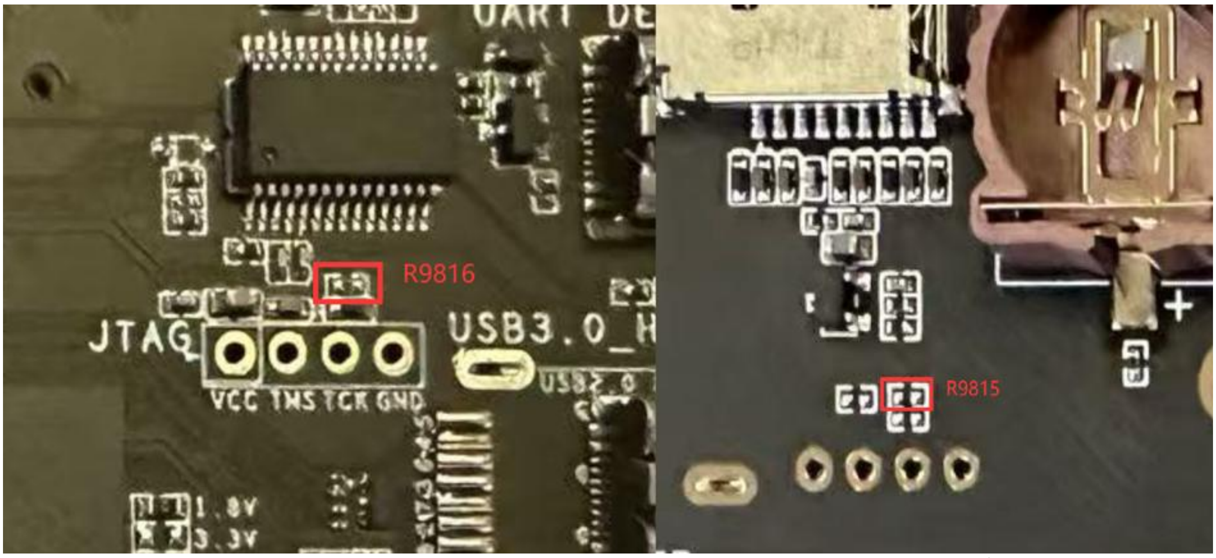


Figure 3-9 RV1126B EVB1 JTAG Connector

3.9. MIPI Input Interface

The MIPI input interface uses a horizontal 40pin socket with a pitch of 0.5mm (specifications are detailed in Section 2.3) and supports two ways MIPI inputs.

- Supports IRCUT switching circuit, which can control the day/night mode of the module.
- Supports two input modes: 2x4lane and 4x2lane.
- The module board reserves MCLK and MIPI CLK signals, which can be expanded for stereo/quadruple-eye applications as needed.

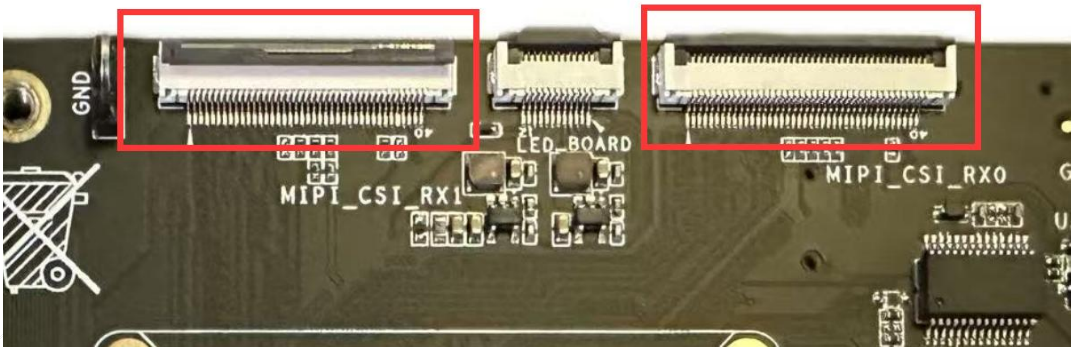


Figure 3-10 RV1126B EVB1 MIPI Input Interface

The signal sequence of the MIPI RX interface is as follows:

No.	Signal	IO Voltage	No.	Signal	IO Voltage
1	GND		21	MIPI_PDN0	1.8V
2	MIPI_D0N		22	NC	
3	MIPI_D0P		23	I2C_SCL	1.8V

No.	Signal	IO Voltage	No.	Signal	IO Voltage
4	GND		24	I2C_SDA	1.8V
5	MIPI_D1N		25	SPI_MISO	3.3V
6	MIPI_D1P		26	SPI_CS	3.3V
7	GND		27	GND	
8	MIPI_CLK0N		28	3V3	
9	MIPI_CLK0P		29	3V3	
10	GND		30	3V3	
11	MIPI_D2N		31	GND	
12	MIPI_D2P		32	IRCUT_A/SPI_CLK	3.3V
13	GND		33	IRCUT_B/SPI_MOSI	3.3V
14	MIPI_D3N		34	MIPI_RST1	1.8V
15	MIPI_D3P		35	GND	
16	GND		36	MIPI_MCLK1	1.8V
17	MIPI_CLK0	1.8V	37	GND	
18	GND		38	MIPI_CLK1N	
19	MIPI_PDN1	1.8V	39	MIPI_CLK1P	
20	MIPI_RST0	1.8V	40	GND	

3.10. LED Extboard Interface

The LED interface uses a horizontal 12pin socket with a pitch of 0.5mm (specifications are detailed in Section 2.3) for controlling the light board display, photoresistor input, and ambient light detection input.

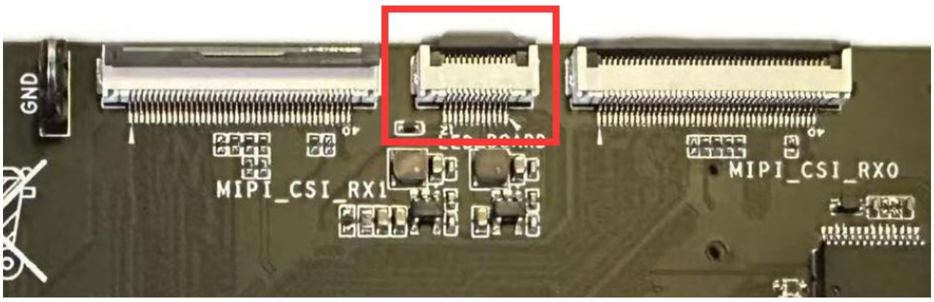


Figure 3-11 RV1126B EVB1 LED Interface

No.	Signal	IO Voltage	No.	Signal	IO Voltage
1	SENSOR_INT	3.3V	7	VCC_IR_LEDK	
2	I2C2_SDA_SENSOR	3.3V	8	VCC_W_LEDK	
3	I2C2_SCL_SENSOR	3.3V	9	GND	
4	GND		10	VCC3V3_SENSOR	

No.	Signal	IO Voltage	No.	Signal	IO Voltage
5	VCC_W_LEDA		11	NC	
6	VCC_IR_LEDA		12	SARADC0_IN1_CDS	1.8V

3.11. Display Output Interface

The EVB1 reserves one MIPI DPHY DSI TX interface, which is equipped with a 1080P screen by default. It supports users to expand the LCD display.

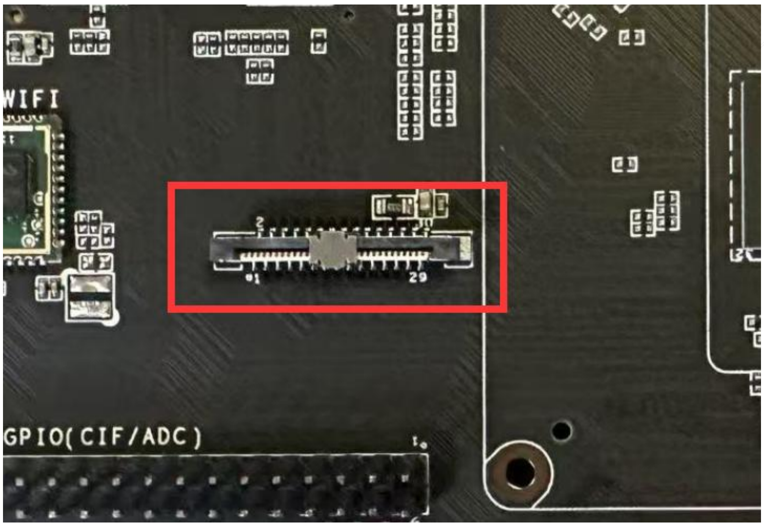


Figure 3-12 RV1126B EVB1 display output interface

The signal sequence of the MIPI output interface is as follows (please refer to the schematic diagram for the specific IO selection):

No.	Signal	IO Level	No.	Signal	IO Level
1	GND		16	GND	
2	MIPI_D0N		17	LCD_PWM_BL	3.3V
3	MIPI_D0P		18	LCD_RST	NC
4	GND		19	3V3	
5	MIPI_D1N		20	LCD_EN	NC
6	MIPI_D1P		21	LCD_CS	NC
7	GND		22	BL_EN	3.3V
8	MIPI_CLKN		23	I2C_SCL	3.3V
9	MIPI_CLKP		24	I2C_SDA	3.3V
10	GND		25	TP_INT	3.3V
11	MIPI_D2N		26	TP_RST	3.3V
12	MIPI_D2P		27	GND	

No.	Signal	IO Level	No.	Signal	IO Level
13	GND		28	5V0_1	
14	MIPI_D3N		29	5V0_2	
15	MIPI_D3P		30	5V0_3	

3.12. USB2.0 Interface

The development board supports one USB2.0 DEVICE interface. The interface is a standard TYPEC port, facilitating firmware programming and ADB debugging for users.

If the USB3.0_HOST function is required, you can replace the TYPEC interface with a TYPEA interface.

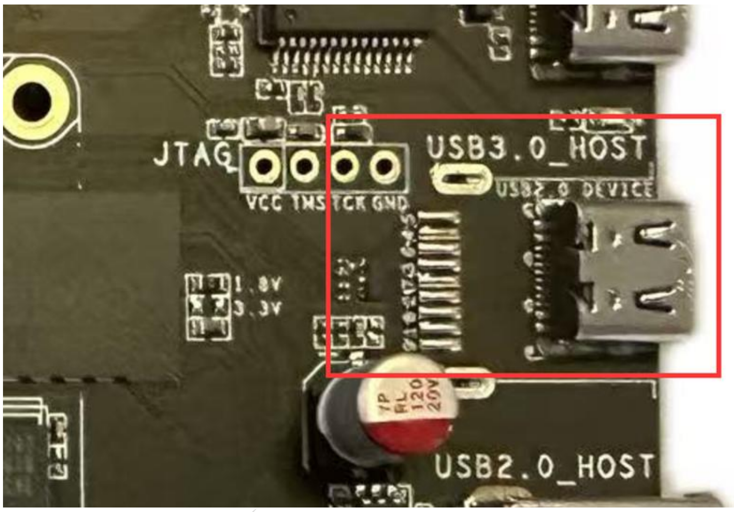


Figure 3-13 RV1126B EVB1 USB2.0 Interface

3.13. CIF/ADC Daughter Board Interface

The development board has reserved a VI_CIF/ADC 2.54mm pin header, which can be used as a CIF module input or for GPIO functions. You can make corresponding convert boards according to your needs.

The VI_CIF interface has a default level of 3.3V. If a 1.8V interface level is required, you can change R1623 to R1622, as shown in the figure below.

Figure 3-14 RV1126B CIF/ADC interface

The interface signal sequence as follow:

No.	Signal	IO Level	No.	Signal	IO Level
1	ADC_IN3	1.8V	2	ADC_IN4	1.8V
3	ADC_IN5	1.8V	4	ADC_IN6	1.8V
5	CIF_RST	1.8V	6	CIF_PDN0	1.8V
7	I2C3_SDA_CIF	1.8V	8	I2C3_SCL_CIF	1.8V
9	GND		10	GND	
11	CIF_CLKOUT_M0	3.3V	12	CIF_CLKIN_M0	3.3V
13	GND		14	GND	
15	CIF_D12_M0	3.3V	16	CIF_D14_M0	3.3V
17	CIF_D0_M0	3.3V	18	CIF_D10_M0	3.3V
19	CIF_D6_M0	3.3V	20	CIF_D8_M0	3.3V
21	CIF_D15_M0	3.3V	22	CIF_D4_M0	3.3V
23	CIF_D11_M0	3.3V	24	CIF_D13_M0	3.3V
25	CIF_VSYNC_M0	3.3V	26	CIF_D9_M0	3.3V
27	CIF_D7_M0	3.3V	28	CIF_HSYNC_M0	3.3V
29	CIF_D5_M0	3.3V	30	CIF_D2_M0	3.3V
31	CIF_D1_M0	3.3V	32	CIF_D3_M0	3.3V
33	GND		34	GND	
35	GND		36	GND	
37	VCC5V0_CIF		38	VCC3V3_CIF	
39	VCC5V0_CIF		40	VCC3V3_CIF	

3.14. RGMII/CIF Daughter Board Interface

The RGMII/VI_CIF interface reserved on the development board is led out using the 61082-061402LF connector. It can be used as the input of the CIF module, for gigabit network port conversion, and for GPIO functions. Corresponding adapter boards can be made according to requirements.

The RGMII interface has a default level of 3.3V. If a 1.8V interface level is required, change R1734 to R1733, as shown in the figure below.

The modification of the 1.8V level for the CIF part signals is the same as in the previous section.

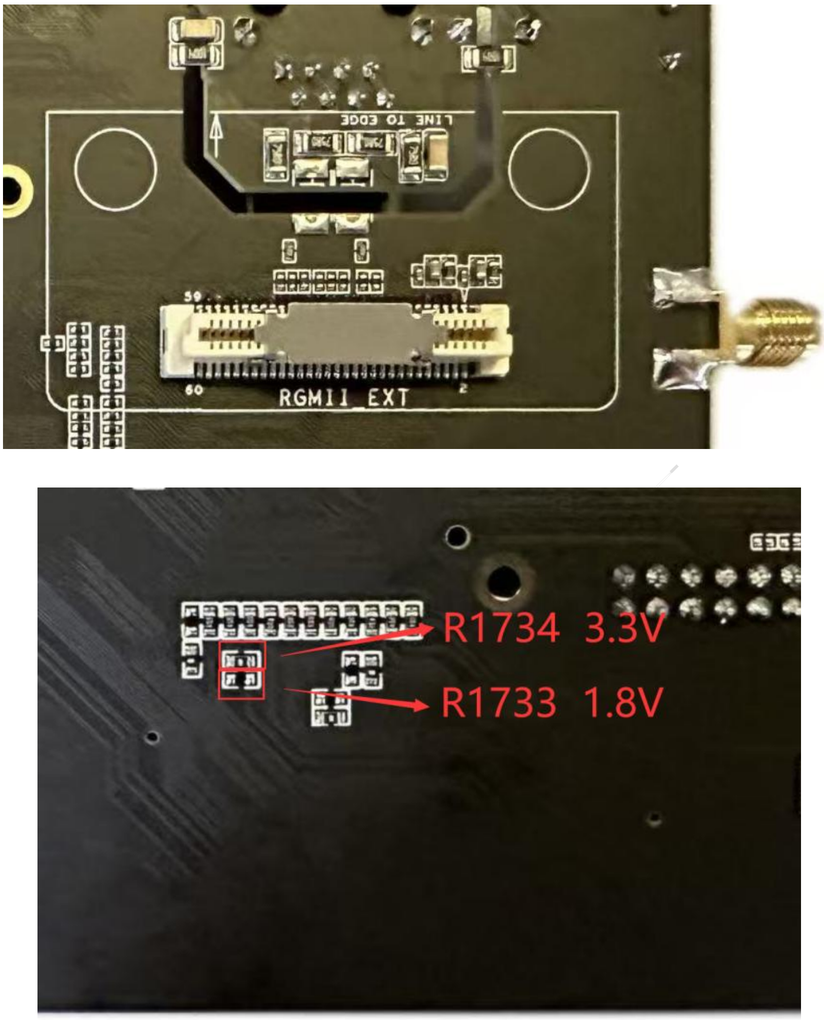


Figure 3-15 RV1126B EVB1 RGMII/CIF interface

The interface signal sequence as follow:

No.	Signal	IO Level	No.	Signal	IO Level
1	VCC3V3_ETH		2	VCC5V0_ETH	
3	VCC3V3_ETH		4	GND	
5	CON_PIN5/ VI_IRFPA_D1	3.3V	6	VI_IRFPA_D3	3.3V
7	CON_PIN7/VI_IRFPA_D2	3.3V	8	VI_IRFPA_D4	3.3V
9	VCCIO_ETH	3.3V	10	ETH_RST/IRFPA_RST	3.3V
11	GND		12	ETH_INT/VO_OCC_SDA1	3.3V
13	ETH_PPS/IRFPA_3.6V_PWREN	3.3V	14	CON_PIN14/VI_IRFPA_D0	3.3V
15	ETH_PTP_REFCLK	3.3V	16	VI_IRFPA_D5	3.3V

No.	Signal	IO Level	No.	Signal	IO Level
17	CLK_OUT_ETH/VO_OCC_SDA5	3.3V	18	VI_IRFPA_D6	3.3V
19	GND		20	VI_IRFPA_D7	3.3V
21	ETH_MCLK	3.3V	22	VI_IRFPA_D8	3.3V
23	GND		24	VI_IRFPA_D9	3.3V
25	ETH_RXCLK	3.3V	26	VI_IRFPA_D10	3.3V
27	GND		28	VI_IRFPA_D11	3.3V
29	ETH_RXCTL/VO_OCC_SDA2	3.3V	30	VI_IRFPA_D12	3.3V
31	ETH_RXD0/VO_OCC_SDA3	3.3V	32	VI_IRFPA_D13	3.3V
33	ETH_RXD1	3.3V	34	GND	
35	ETH_RXD2	3.3V	36	ETH_LED_SPD/IRFPA_MCLK	3.3V
37	ETH_RXD3	3.3V	38	GND	
39	GND		40	VI_IRFPA_CLKIN	3.3V
41	ETH_TXCLK	3.3V	42	IRCUT_A1/IRFPA_SHUTTER+	3.3V
43	GND		44	IRCUT_B1/IRFPA_SHUTTER-	3.3V
45	ETH_TXCTL/VO_OCC_SDA6	3.3V	46	I2C3_SCL_IRFPA	1.8V
47	ETH_TXD0	3.3V	48	I2C3_SDA_IRFPA	1.8V
49	ETH_TXD1/VO_OCC_SDA4	3.3V	50	GND	
51	ETH_TXD2	3.3V	52	ETH_PPSCLK/IRFPA _PWREN	3.3V
53	ETH_TXD3/VO_OCC_SDA0	3.3V	54	VI_IRFPA_VSYNC	3.3V
55	GND		56	VI_IRFPA_HSYNC	3.3V
57	ETH_MDIO	3.3V	58	ETH_LED_ACT/VO_OCC_FSYNC	3.3V
59	ETH_MDC	3.3V	60	GND	

3.15. TF Card Interface

The development board supports one TF card interface, compatible with SDMMC2.0 and SDMMC3.0 modes.

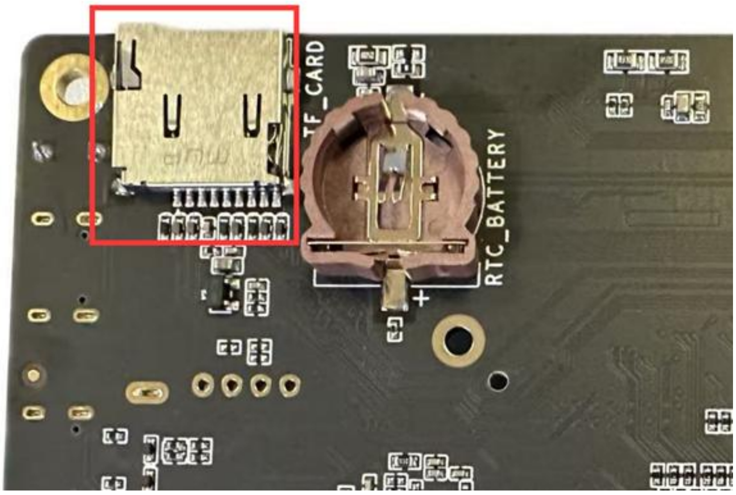


Figure 3-16 RV1126B EVB1 TF Card Interface

3.16. Audio Input

The development board supports two differential mic input. By default, the on-board silicon mic is used.

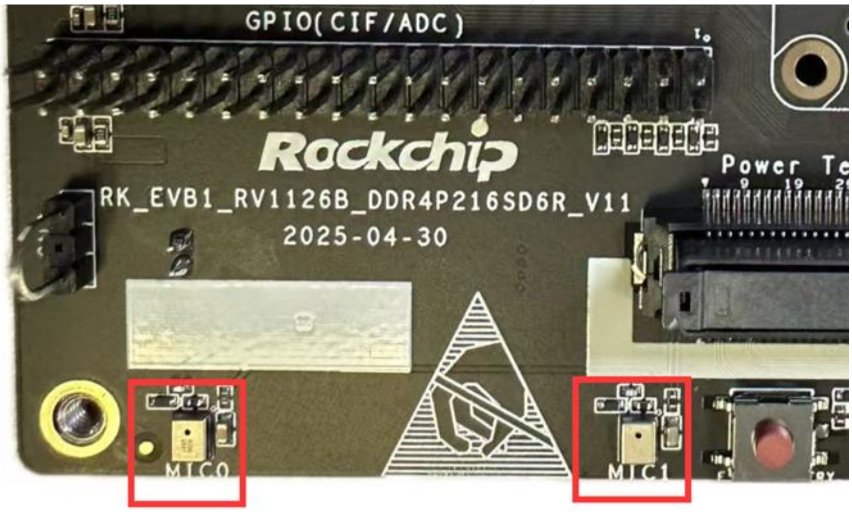


Figure 3-17 RV1126B EVB1 Audio Input

3.17. Audio Output

The development board reserves one Speaker interface, supporting a maximum output power of 3W (10% THD, 4ohm speaker).

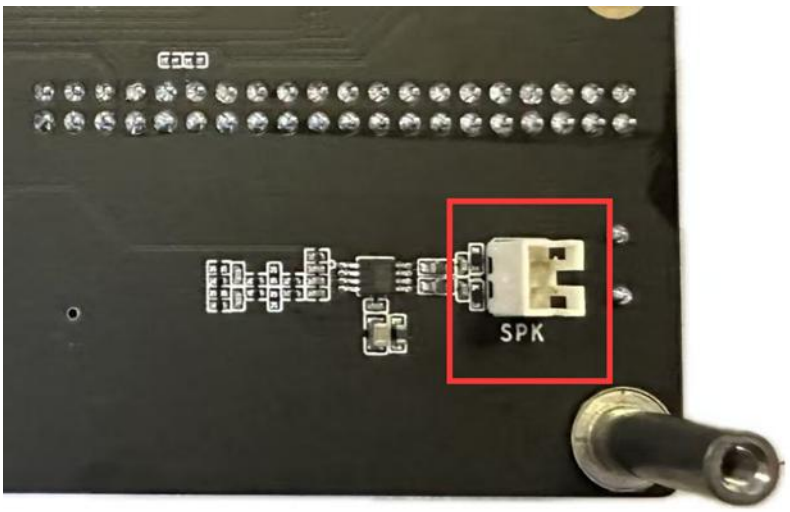


Figure 3-18 RV1126B EVB1 Speaker Connector

4. Getting Started Tutorial

4.1. Powering On and Off

The power-on and power-off methods for the RV1126B EVB1 are as follows:

- Power-on method

Use DC 12V/3A for power supply, and turn on the power switch. Serial port output information indicates that the default firmware has started successfully.

- Power-off method:

Turn off the power switch.

4.2. Debugging Methods

4.2.1. Serial Debugging

Connect the USB TYPEC Debug interface on the right side of the development board to the PC, and obtain the current COM port number in the Device Manager on the PC.

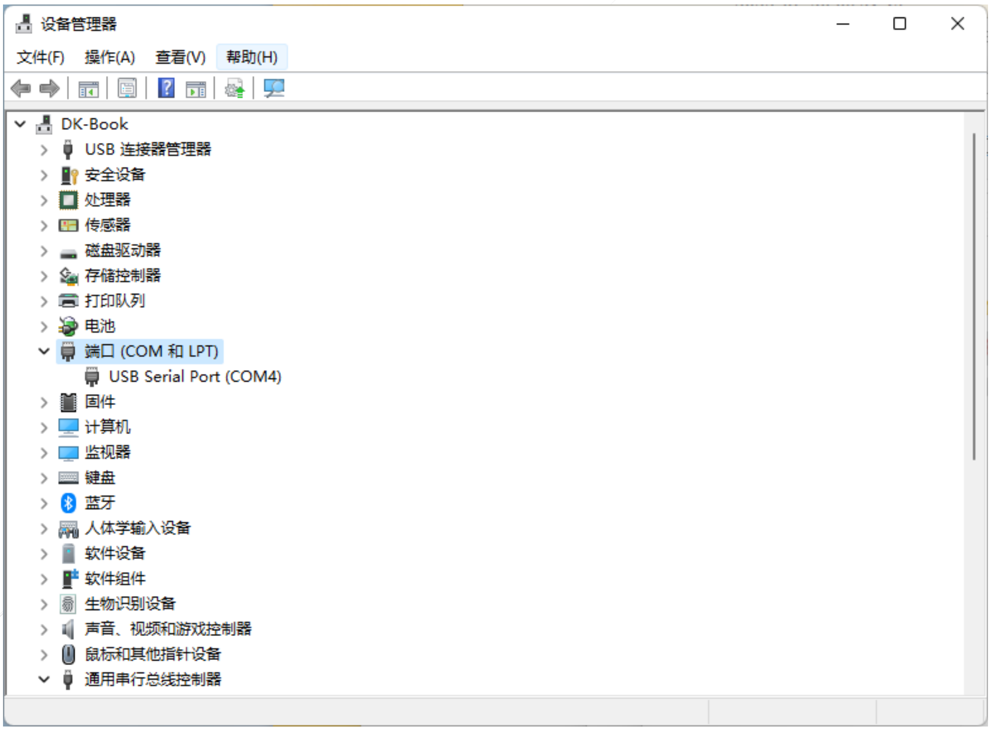


Figure 4-1 Obtaining the Current COM Port Number

Open the serial port tool, configure the serial port tool according to Table 4-1, and connect the serial port after the settings are completed to enter the serial debugging interface. Correct characters on the interface indicate a successful serial connection, and you can then use the serial port for debugging.

Table 4-1 Serial Configuration Information

Configuration Item	Parameter
Port	COM port number obtained from Device Manager
Baud rate	1500000
Data bit	8
Parity	None
Stop bit	1
Flow control	None

Note:

Due to the limitations of the conversion chip, it’s need to connect the USB interface of the serial port first, and then turn on the main power switch. Otherwise, there may be no log output.

4.2.2. ADB Debugging

(1) Ensure that the driver installation in Section 4.3.1 is successful, and connect the USB2.0 TYPEC interface of EVB1 to PC;

(2) Power on the development board and wait for the system to boot;

(3) Open the adb tool ;

(4) Type "adb shell" to enter the adb debugging window.

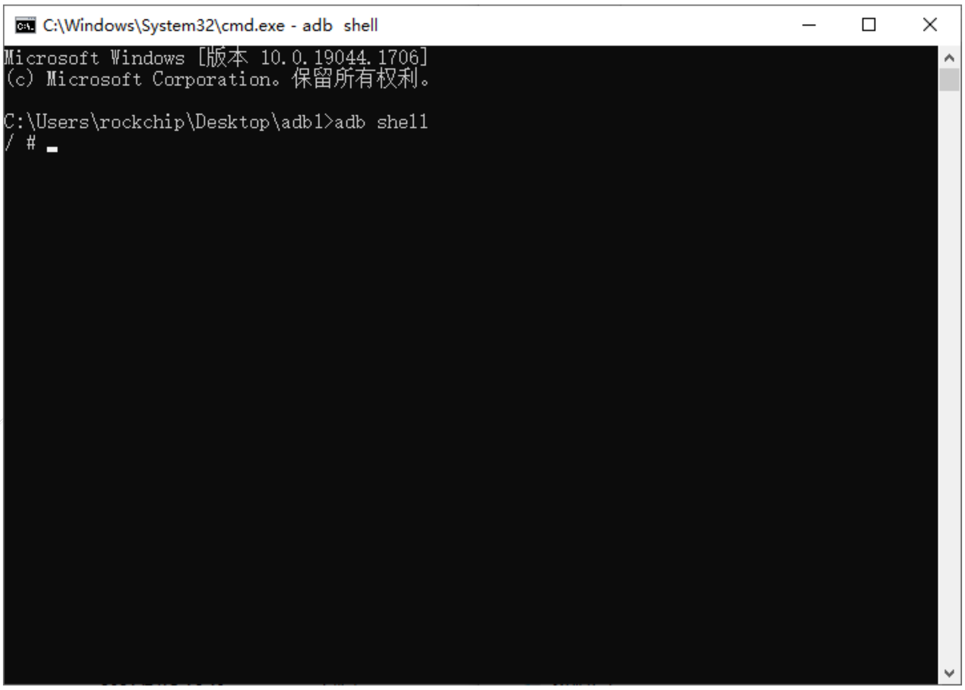


Figure 4-2 ADB Debugging Window

4.3. Firmware Download and Upgrade

RV1126B EVB1 supports USB, UART, TFTP, Micro-SD and others firmware updating method.

4.3.1. Driver Installation

The driver needs to be installed before the EVB1 driver upgrading. The following describes the driver installation process under Windows system.

Find **DriverAssitant_v5.12** in the provided tool folder, and click **DriverInstall.exe** to open the installation screen. Click "Install Driver" and wait for the successful installation prompt. If the old driver has been installed, please click "Uninstall Driver " and reinstall the driver.

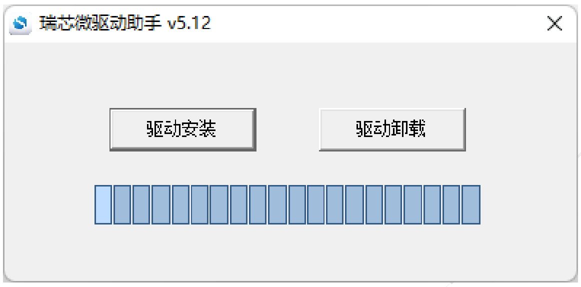


Figure 4-3 Drive Assistant installation screen

4.3.2. USB Update Mode

The specific steps are as follows:

1) Connect the DEVICE port of USB TYPEC to the PC side and keep pressing the MASKROM key of EVB1.

2) Turn on the 12V power supply switch of EVB1. If EVB1 is already powered on, press the reset key.

3) Release the key after the upgrade tool shows that a MASKROM device is detected.

4) The download tool supports discrete firmware download and packaged firmware download. The following are two methods for selecting firmware:

a) Open the tool, select RV1126B, enter the interface as shown in Figure 4-4, select "Search Path..." → select the discrete firmware folder "IMAGES" → click "Select Folder" - check the discrete files to be downloaded → click the "Download" button.

b) Open the tool, select RV1126B, enter the interface as shown in Figure 4-5, select "Firmware..." → select the packaged firmware folder "IMAGES" → select the update file → click the "Upgrade" button.

5) Click "Download/Upgrade" to enter the upgrade state. The right side of the tool is the progress display bar, which shows the download progress and verification status.

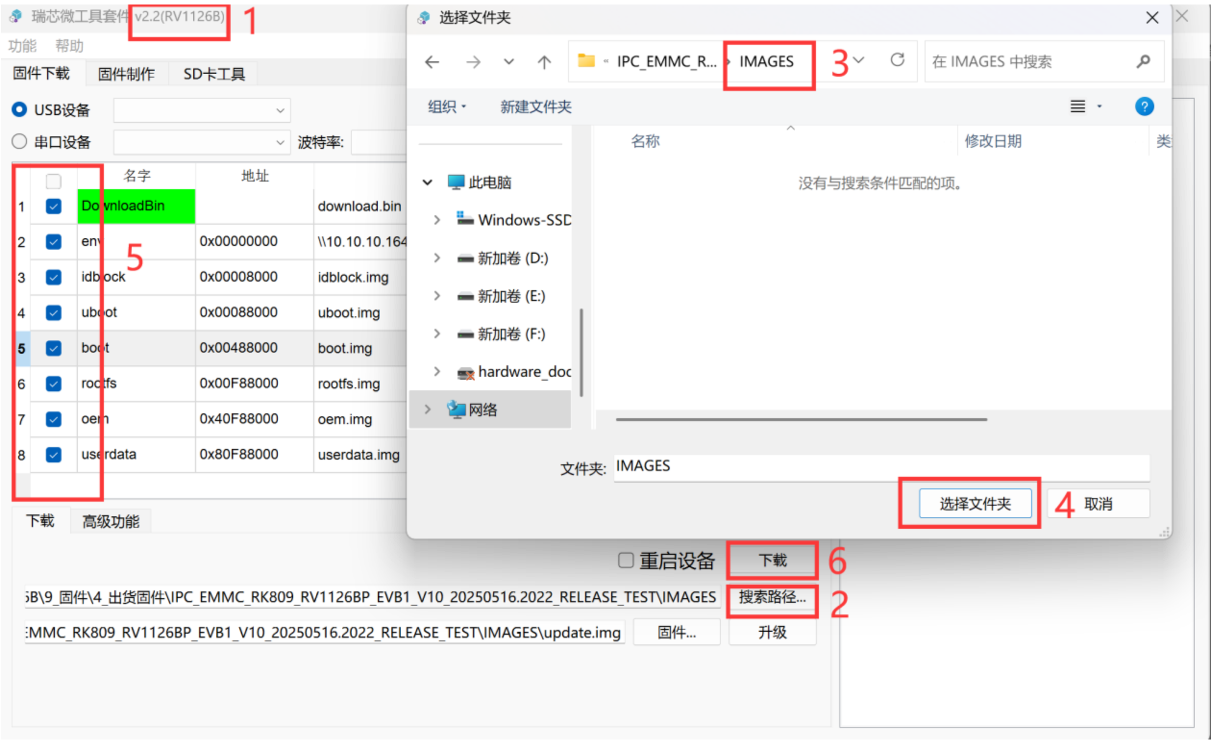


Figure 4-4 Discrete firmware

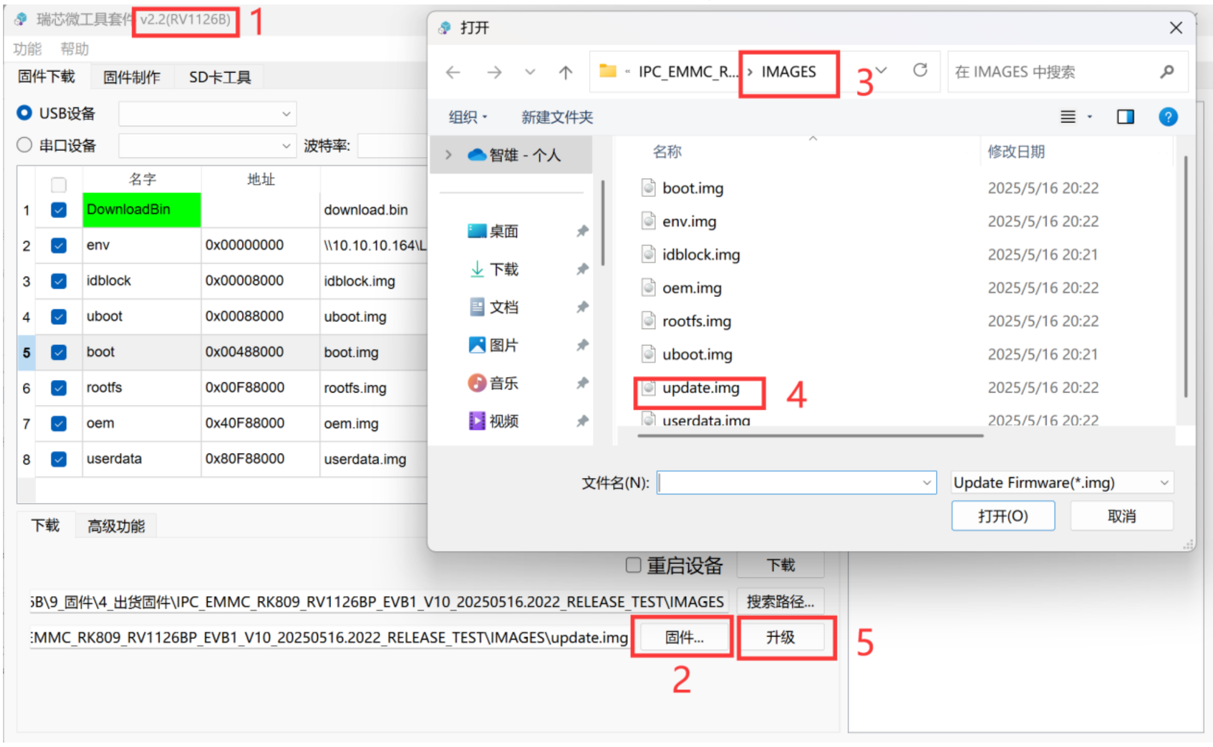


Figure 4-5 Packaged Firmware

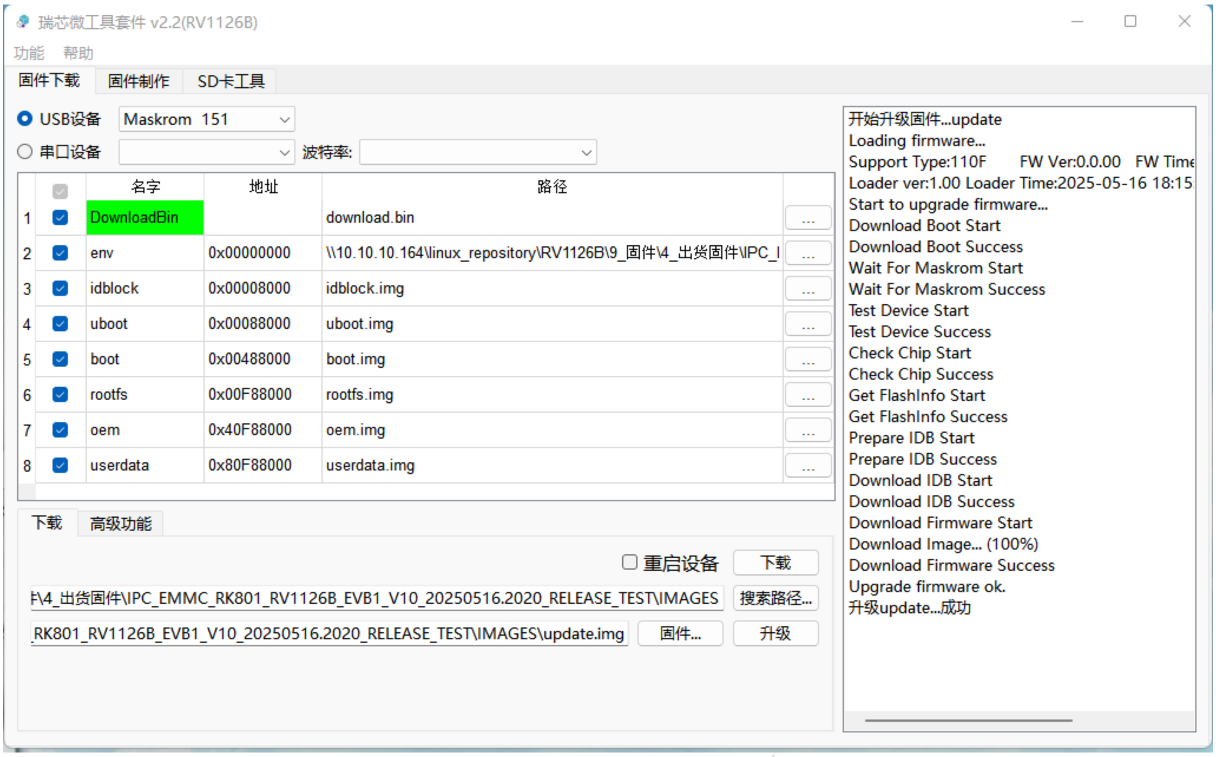


Figure 4-6 USB upgrade mode

4.3.3. UART Update Method

The specific steps are as follows:

1) Connect the UART DEBUG port of TYPE-C to the PC side (disconnect from the PC-side serial port tool), and keep pressing the MASKROM button of EVB1.

2) Turn on the 12V power supply switch of EVB1; if EVB1 is already powered on, press the reset key.

3) After the upgrade tool shows that a COM device is detected, release the key. Select the corresponding baud rate of the serial port. By default, two options of "115200" and "1500000" are available.

4) Select the firmware as described above.

5) Click "Download/Upgrade" to enter the upgrade state. The right side of the tool is the progress display bar, showing the download progress and verification status.

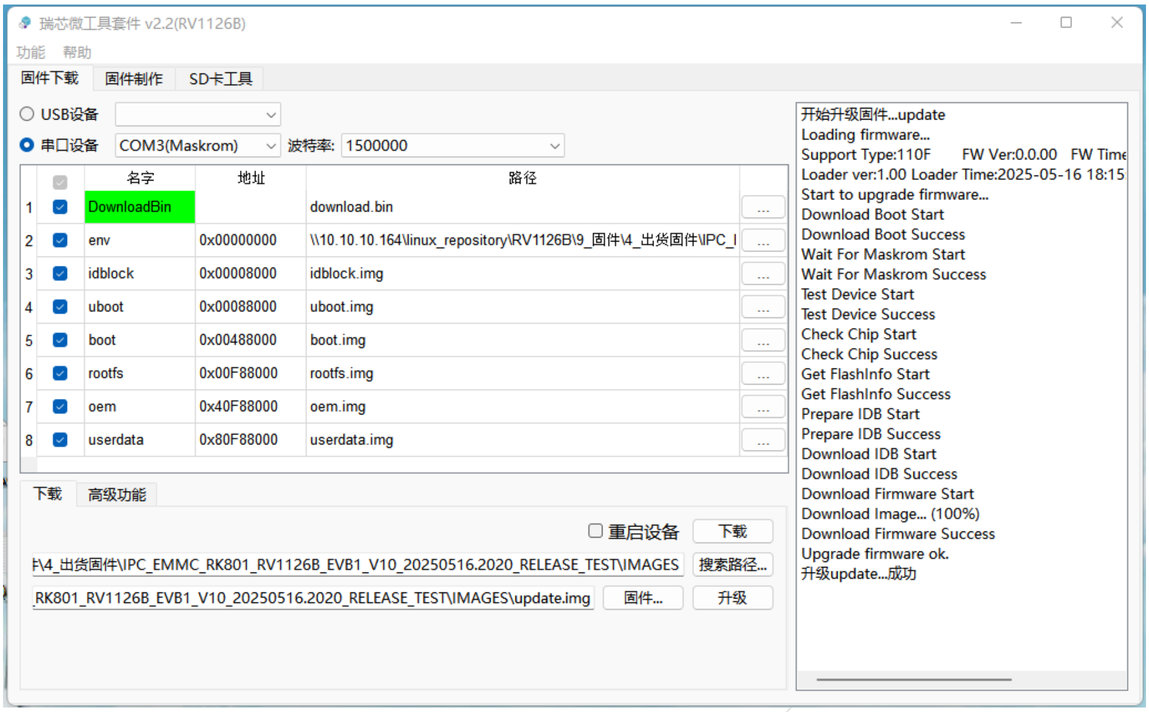


Figure 4-7 UART update method diagram

4.3.4. TFTP Update Method

TFTP upgrade files will be compiled with the firmware in the “<sdk>/output/image/” directory. The file is named *tftp_update.txt*. Usage is as follows:

- Config the TFTP servers

TFTPD64 download address: https://pjo2.github.io/tftpd64

Note:

1) Use Tftpd64 software need to comply with relevant open source protocols.
2) All legal risks and other consequences that the Tftpd64 may bring is borne by the user.

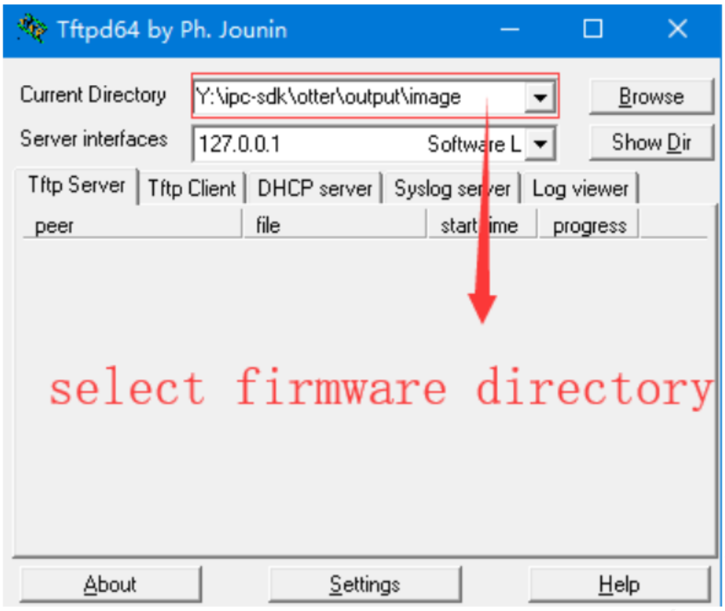


Figure 4-8 TFTP update mode

- Put the update file tftp_update.txt and all firmware that suffixes named .img in the directory specified in the server (Note: SLC NAND does not support the idblock partition updating by download firmware.)
- Set the IP address on the U-Boot terminal (the IP address in the follow figure is for reference only. Please set it by yourself according to the actual scene to ensure that the client and the server are in the same network segment.)

```
=> setenv ipaddr 192.168.1.111
=> setenv serverip 192.168.1.100
=> saveenv
Saving Environment to envf...
=>
```

Figure 4-9 Setting the IP address on the U-Boot terminal

- Run “tftp_update” on the U-Boot terminal

```
=>  tftp_update
ethernet@ffc40000 Waiting for PHY auto negotiation to complete. done
Using ethernet@ffc40000 device
TFTP from server 192.168.1.100; our IP address is 192.168.1.111
Filename 'tftp_update.txt'.
Load address: 0x3be24c00
Loading: *•#
     203.1 KiB/s
done
Bytes transferred = 1250 (4e2 hex)
...
```

Figure 4-10 running the upgrade command on the U-Boot terminal

5. Precautions

The RV1126B EVB1 is designed for laboratory or engineering environments. Please read the following precautions before starting any operations:

- Hot plugging of the display interface and expansion boards is strictly prohibited under any circumstances.
- Before unboxing the development board and proceeding with installation, take necessary anti-static measures to prevent electrostatic discharge (ESD) from damaging the hardware of the development board.
- When handling the development board, hold it by the edges. Avoid touching the exposed metal parts on the board to prevent static electricity from damaging the components.
- Place the development board on a dry, flat surface to ensure it is kept away from heat sources, sources of electromagnetic interference and radiation, and devices sensitive to electromagnetic radiation (such as medical equipment).